

YaP-Folia soak & knob profiles

Operator guide for YaP-Folia performance knobs and soak gates. Patch inventory:

[vendor/folia/patches/AGENT3.md](#) · [AGENT4.md](#) .

Product defaults (ship) — what we cite

Knob	Default	Why
<code>folia-async-chunk-save</code>	true	Moonrise flush off region thread
<code>folia-hopper-tick-budget</code>	64	Soft-defer excess hopper transfers per region tick
<code>folia-scoreboard-swmr</code>	true	Correctness under Folia
<code>folia-teleport-transactions</code>	true	Cross-region TP integrity
<code>folia-entity-tick-budget</code>	400	MSPT-gated Mob AI cap (never players/TNT/vehicles/items/bosses)
<code>folia-budget-mspt-threshold</code>	12	Shared gate for entity budget + microtick
<code>folia-entity-tick-max-deferred</code>	40	Anti-starve: force-tick after N consecutive skips
<code>folia-microtick-budget-ms</code>	8	Soft Mob AI deadline on hot regions
<code>folia-subregion-partition</code>	true	Parallel shards when hot + geometry allows
<code>folia-subregion-mspt-clear</code>	16	Hysteresis vs engage threshold (20)
<code>folia-subregion-coalesce-min-wall-ms</code>	30000	Anti-thrash before coalesce

Official cites use this ship profile. Stock Folia / Canvas ignore YaP `-D` knobs — that is the product delta. Result JSON records `knob_*` fields; `YAP_MSPT_REQUIRE_SHIP_KNOBS=1` fails the gate if micro/subregion/entity are missing or below ship floor.

Scheduler: `folia-kernel/config/paper-global.yml` → `threaded-regions.scheduler: WORK_STEALING` .

Soak ladder

```
./scripts/yapctl soak-compatible      # boot + API (~5–15 min) – partition/budget ON
./scripts/yapctl soak-perf 30         # heap/thread samples
./scripts/yapctl soak-long 12         # 12h default, 8h floor
./scripts/yapctl cite-fullcite        # stock Folia vs YaPcore ≥5% with **ship knobs**
```

Profile: ship cite (default)

```
# cite-fullcite.sh already exports ship knobs + NO_DIG + VD/sim 8
./scripts/yapctl cite-fullcite

# Heavypop + Canvas peer:
YAP_BENCH_COMPETITORS=folia,canvas,yapcore YAP_BENCH_SHUFFLE=0 \
YAP_MSPT_REQUIRE_SHIP_KNOBS=1 \
./scripts/bench/run-vs-folia.sh heavypop 40
```

Profile: baseline A/B (prove knobs help)

```
# Async+hopper only – no smart budget / microtick / partition
YAP_BENCH_CITE_BASELINE=1 \
YAP_FOLIA_ASYNC_CHUNK_SAVE=true YAP_FOLIA_HOPPER_TICK_BUDGET=64 \
./scripts/bench/run-vs-folia.sh heavypop 40
```

Compare baseline vs ship on the same machine/stamp pair.

Profile: spawn-hot (aggressive — lab)

```
YAP_FOLIA_ENTITY_TICK_BUDGET=300 \
YAP_FOLIA_MICROTICK_BUDGET_MS=8 \
YAP_BENCH_FULL_STACK=1 \
./scripts/bench/run-vs-folia.sh spawncollapse 40
```

Gaming blacklist

- Softening the 5% cite gate or MSPT<2 noise rules
- Physics-off / keepalive-only bots as a “population” cite
- YaP-only lobe layouts vs stock contiguous without fair-paired disclosure
- Claiming chassis JVM time as game-tick MSPT
- Hiding ship knobs — always disclose `knob_*` in JSON / whitepaper
- Claiming “vanilla Folia knobs” when ship smart budget / partition are on